

Advancing monitoring and prediction of antimicrobial resistance trajectories using flexible spatiotemporal modelling: A roadmap for early warning systems

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AMR Data Challenge Finalists

Background and context

Surveillance is essential for tracking antimicrobial-resistance (AMR) **emergence**, virulent strains, and *de novo* mutations.

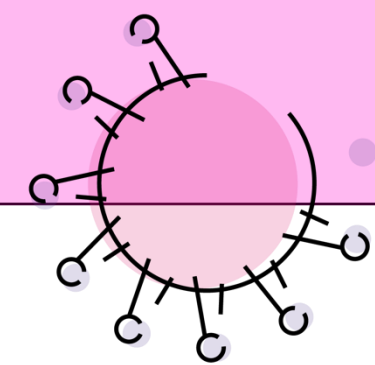
- **Early detection** and control of AMR **shifts** is crucial for public health interventions and resource allocation.

Current statistical methods for change point detection

- Focus on **mean level shifts** in AMR rates (e.g., ARIMA).¹
- Limited in detecting **gradual shifts**.

Advanced detection methods:

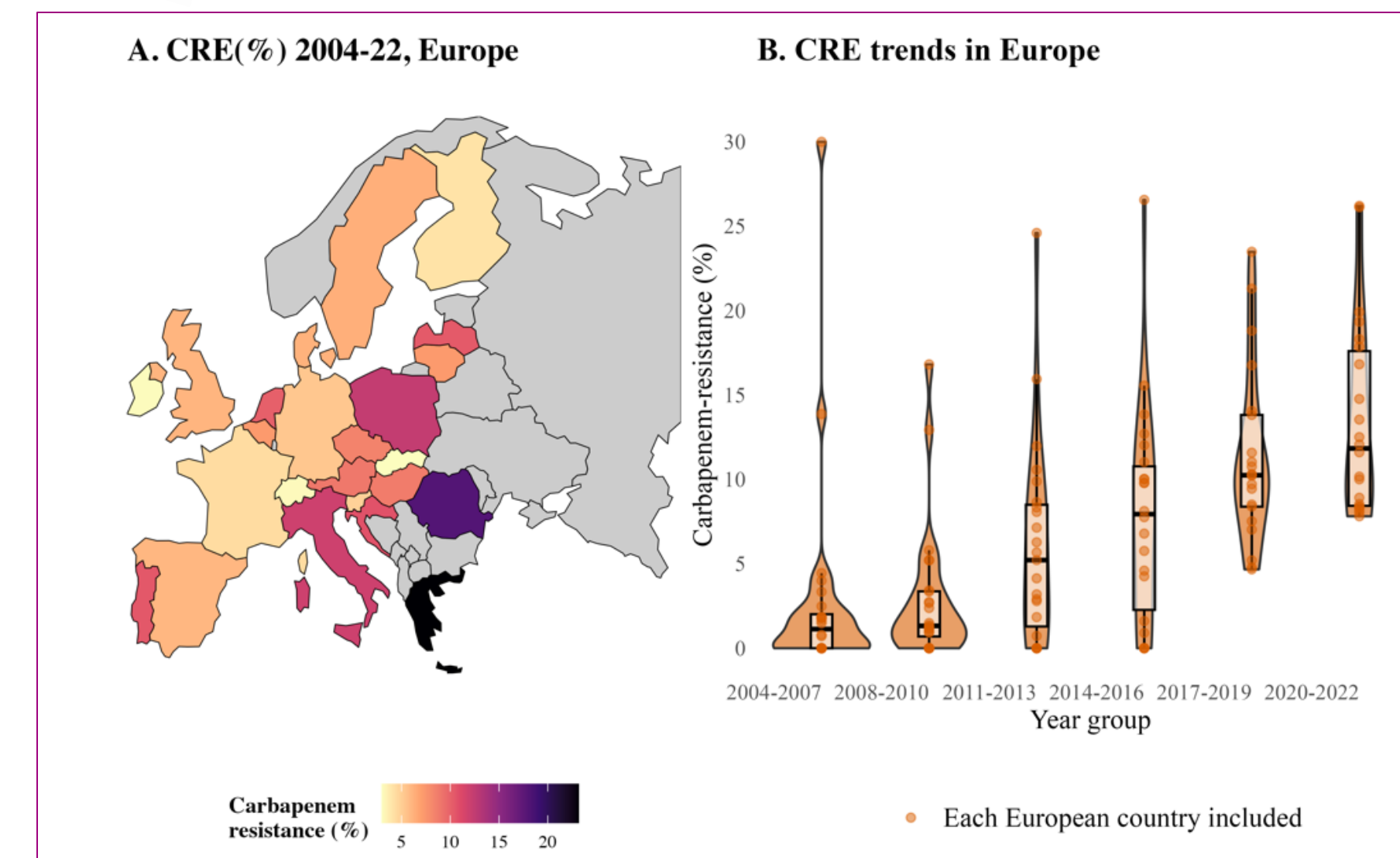
Generalised Additive Models (GAMs): Capture gradual trend changes and provide **earlier risk prediction**.



Study objective: To employ GAMs to identify critical change points in AMR spread over time, using **carbapenem-resistant Enterobacterales (CRE)** in European countries as an exemplar



 **CRE** (Carbapenem-Resistant Enterobacterales)



- CRE prevalence increased from **1.4% in 2004** to **12.3% by 2022** in Europe
- Regional variation: **Greece and Romania** reported the highest CRE prevalence.

¹ Aminikhanghahi, S. and Cook, D.J., 2017. A survey of methods for time series change point detection. Knowledge and information systems, 51(2), pp.339-367.

Objectives & Methods

Data and analytical sample

- Pfizer's **ATLAS** dataset (64,540 bacterial isolates, 24 European countries).
- Analysis of **CRE trends** in bloodstream and urinary tract infections only, **2004-2022**.
- Countries reporting at least four data points over time.



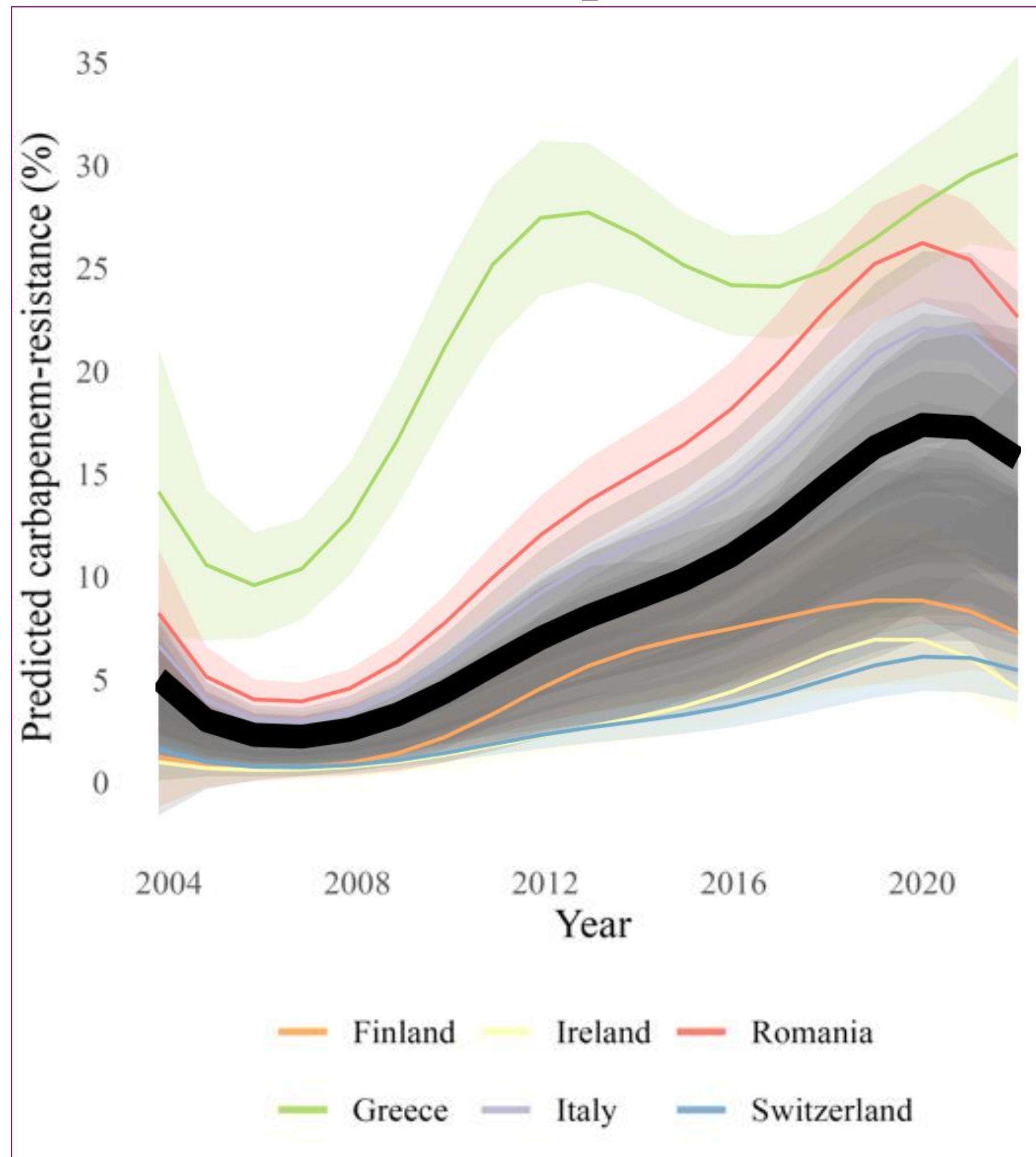
Statistical methods

Primary analyses

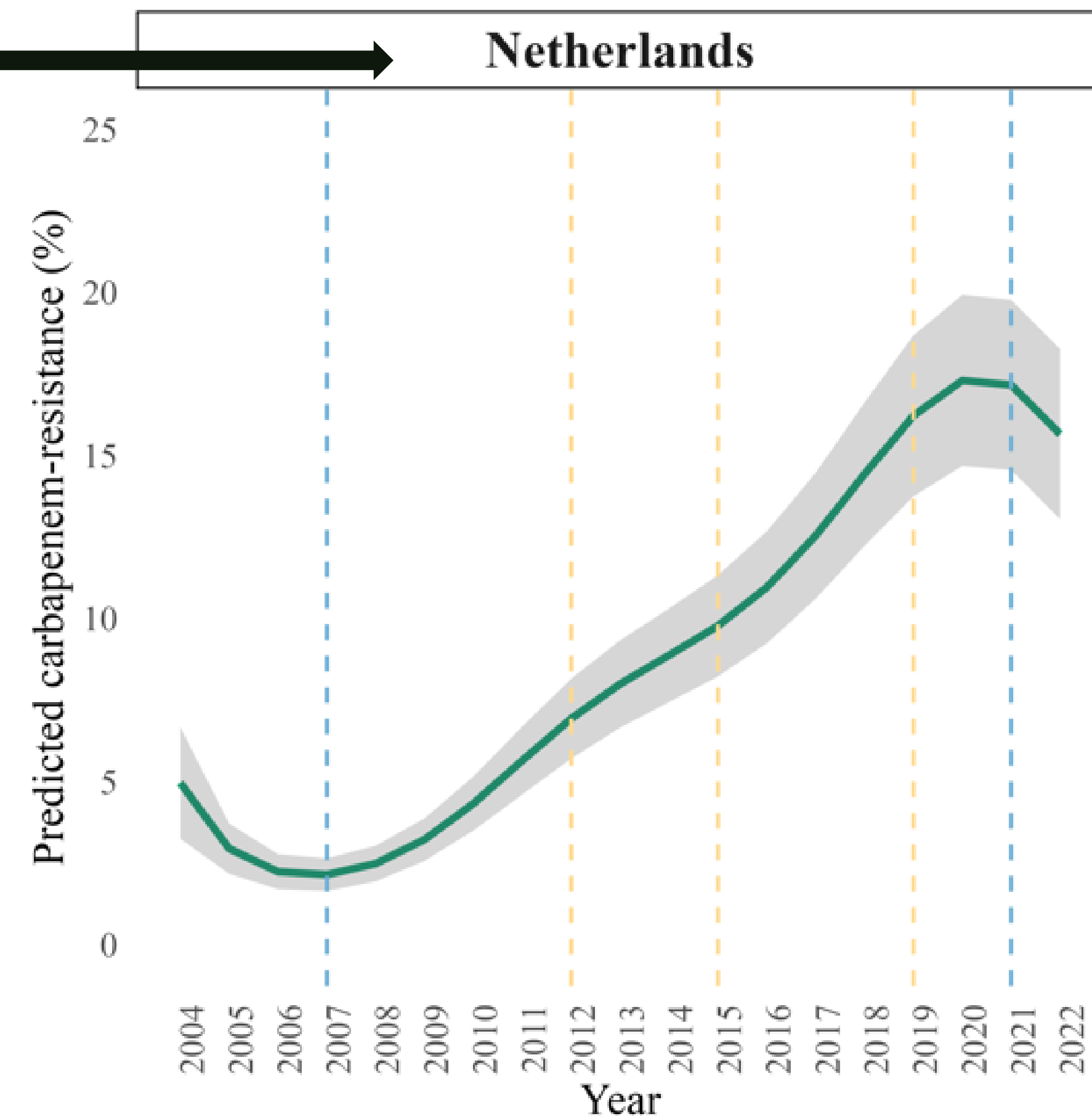
- GAMs with **Gaussian Markov Random Fields** capture spatial interactions between countries, while smooth terms for time (years) are modelled using splines and tensor splines for country-year interactions.
- First and second **derivatives** used to detect **accelerations** and **decelerations** in CRE prevalence over time.
- Calculation of **growth rates** for CRE, using estimated derivatives.

Key findings... let's take an example

Predicted CRE Europe



Predicted CRE, exemplar



Key findings... let's take an example

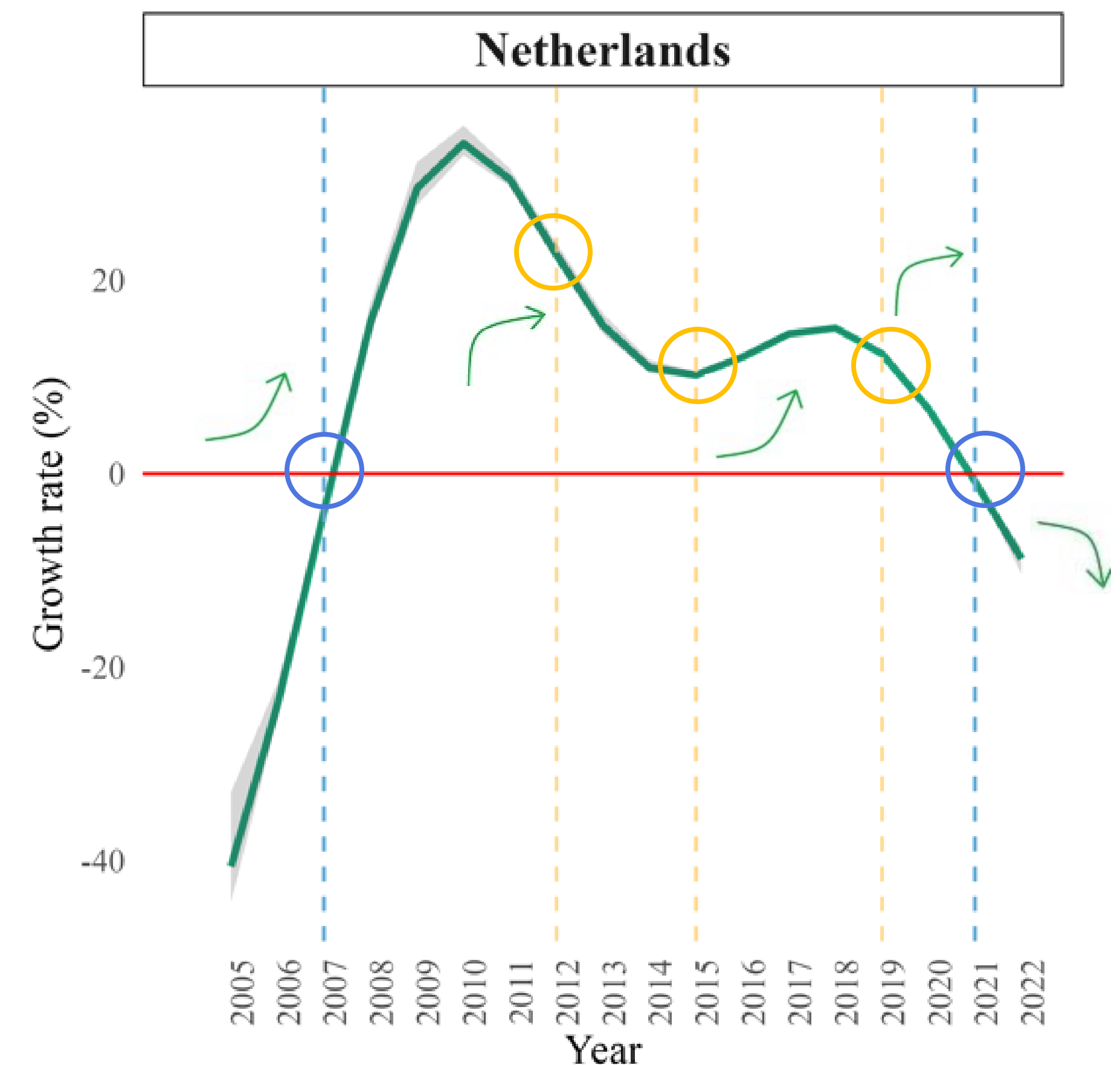
First derivatives (f'): rate of CRE change
Change points ► **First derivatives** if they shift from positive to negative, or vice versa

Second derivatives (f''): Acceleration or deceleration of CRE change
Change-points ► shift in sign and zero fell outside the 95% CI of the **Second derivative**

General Rule Curve Sketching

f'	+	+	-	-
f''	+	-	-	+

Growth rate (CRE), exemplar



--- Change point in first derivative

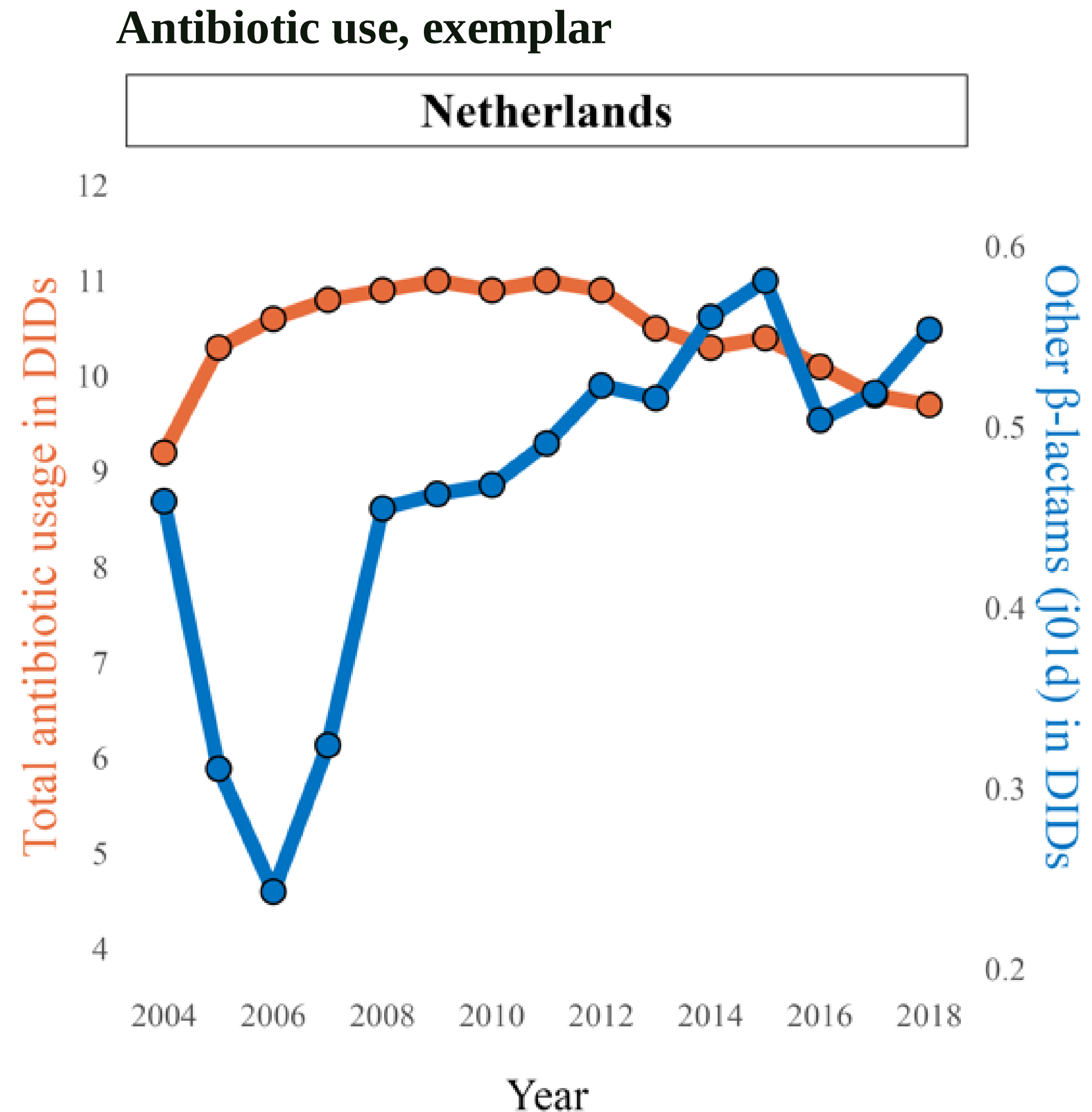
--- Change point in second derivative

Exploratory analysis... let's take an example

‘Exploratory analysis using Global Burden of Disease IHME data’

Increasing trend for **β -lactam** antibiotic usage

Positive correlation (Pearson coefficient 0.45) between **β -lactam** antibiotic usage and CRE prevalence



Implications of the work

What this research adds?

- This framework enables **real-time AMR monitoring**, providing early warnings for **shifting trends**.
- Can guide **resource allocation** and help prioritise **infection control measures** in high-risk regions.
- Supports **AMR National Action Plans** by providing **data-driven insights** for timely interventions.
- **Modelling advantages** of using GAMs: flexibility, spatial relationships, more efficient, +++
- **Broader applicability**: this approach can be scaled to other diseases/pathogens/regions.



Thank you 😊

Questions?

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